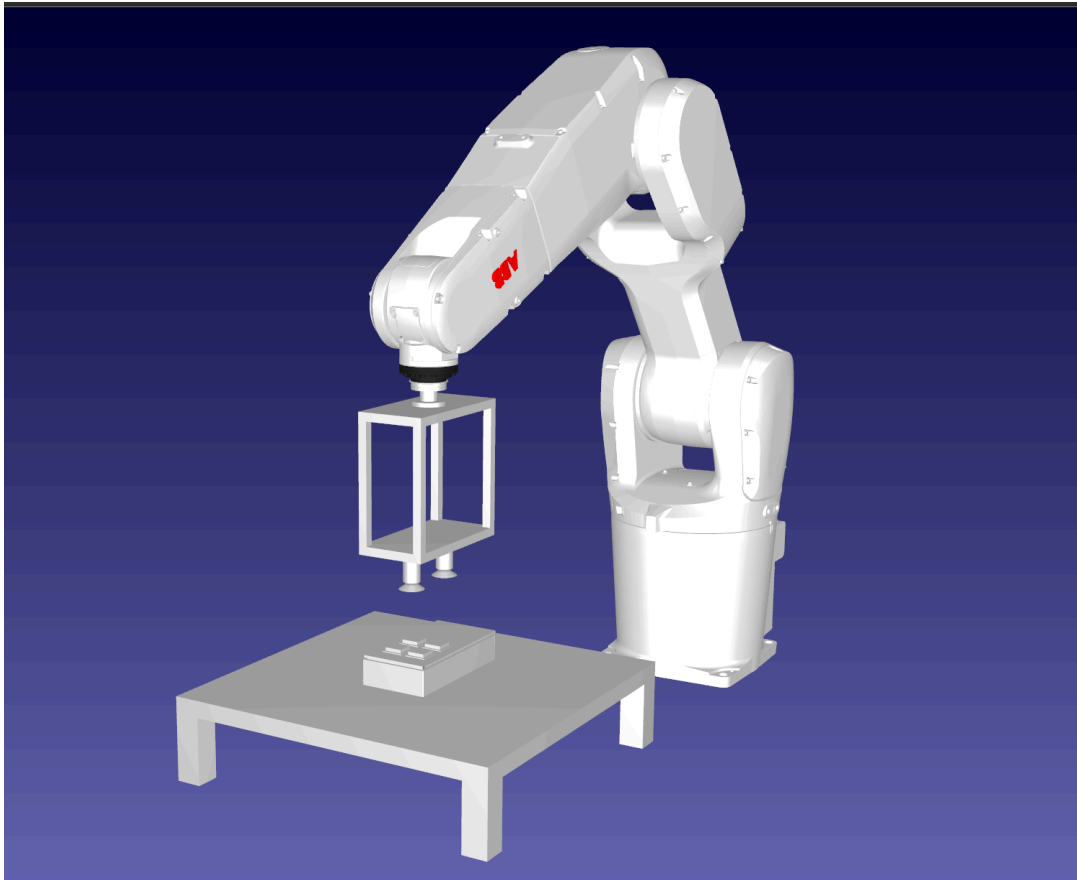


Automated Assembly of a Custom MIDI Pad Controller



MFE 511 Group Project

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1.0 Problem Statement and Design Rationale

The assembly of custom MIDI Pad Controllers presents significant challenges during manual benchtop production. The primary issues include precision, repeatability, production rate, and ergonomic strain. Manual alignment of tactile rubber pads and PCBs often leads to “sticky” buttons or misaligned contacts, which result in a high defect rate. A human operator takes 110s amount of time to complete a unit; to scale production, a 24/7 solution is necessary. Additionally, the repeated pressing of plastic covers and picking small components leads to operator fatigue and inconsistent quality.

1.1 Project Objectives

This project addresses the need for a high-precision, scalable robotic workcell to automate the 4-button MIDI Pad assembly process. The primary goal is to reduce cycle time to an automated assembly benchmark significantly lower than the human rate to justify the capital investment. The system must ensure quality by maintaining placement tolerances within 0.05 mm to guarantee flawless PCB integration and snap-fit fitment. Additionally, the project focuses on scalability by developing a RAPID programming architecture that can be adapted from the current 4-button design to a larger 16-button controller in the future.

1.2 Design Rationale

The development of this automated cell is centered on leveraging the high-performance capabilities of the ABB IRB 1200-5/0.9 to solve the inherent inconsistencies of manual electronic assembly. The rationale for this specific system configuration is built upon three technical pillars: mechanical efficiency, spatial optimization, and process reliability.

1.2.1 Strategic Component Integration

To achieve the target 38-second cycle time, the system utilizes a "no-tool-change" philosophy. By integrating a hybrid EoAT, the robot eliminates the time penalties associated with mechanical tool changers, allowing a seamless transition from the high-precision vacuum handling of PCBs to the high-force mechanical press-fitting of the top acrylic cover. This multi-functional approach ensures that the 5kg payload capacity is used effectively to manage both the tool mass (1.2kg) and the downward forces required for snap-fit engagement.

1.2.2 Workspace and Kinematic Logic

The work-cell layout is mathematically optimized to ensure the robot operates within its peak accuracy range while avoiding kinematic failures. A primary rationale for the cell's geometry is a 45° orientation of the central assembly fixture, a deliberate alignment that would prevent Axis 5 singularities during vertical Z-axis strokes. Such singularities are common in electronics assembly when the robot's wrist is required to remain perfectly perpendicular to the work surface. Furthermore, by utilizing the 900mm reach of the IRB 1200, the design allows for the peripheral placement of gravity feeders and vibratory bowls, ensuring that the primary assembly area remains uncrowded and accessible for high-speed linear movements.

1.2.3 Quality Assurance and ROI

The technical decision to use Smart Component logic and fixed-coordinate RAPID programming provides a cost-effective alternative to expensive vision systems. This ensures a placement repeatability of 0.05mm, which is critical for preventing the "sticky button" defects identified in the problem statement. Ultimately, this design provides a scalable architecture that can transition from a 4-button to a 16-button controller with minimal hardware reconfiguration, maximizing the long-term Return on Investment (ROI).

2.0 System Design Packaging

2.1 Robot Selection and Justification

The ABB IRB 1200-5/0.9 was selected as the primary manipulator for final system implementation due to its critical technical advantages in a production-grade environment. Its 5 kg payload provides a robust safety margin when managing the 1.2 kg hybrid End-of-Arm Tool (EoAT) and the significant downward force required for the final press-fit stage. The robot's 900 mm reach allows for a large work-cell layout, enabling the integration of multiple peripheral feeders without crowding the primary work area. Furthermore, the IRB 1200's advanced motion control allows for high acceleration and deceleration between stations, which is critical for meeting production goals, while its compact footprint and lack of Axis 2 offset minimize the "dead zone" behind the robot.



Fig 1: ABB IRB 1200

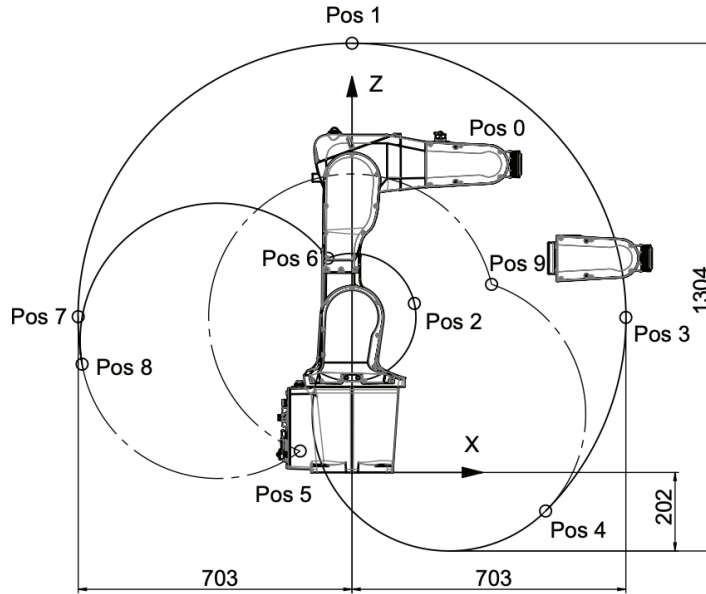


Fig 2: ABB IRB 1200 Working Range

2.2 End-of-Arm Tooling Concept and Design

To maximize operational efficiency, a Custom Hybrid Vacuum/Mechanical Tool has been developed to handle all assembly stages without tool changes within a single work cell. In the RoboDK environment, the hybrid manifold transitions between precision picking and high force pressing by rotating the robot's 6th axis. It features 2 Piab friction suction cups, which are integrated with a vacuum generator to handle the delicate components on the MIDI controller. These suction cups are controlled on the software using the RAPID code, having precise pick and place capabilities for components such as the 160x80mm housing, PCB, and acrylic cover. To ensure that the tool is able to operate within the internal volume of the housing, the cups are mounted with a 15mm standoff to fit the 30mm bottom case and seat internal electronics. The skeletal box design keeps the mechanical tool safely above the rim of the part during the pick and place motion.

For the final sealing stage of snapping the top cover into the case, the robot reorients its 6th axis, rotating the tool 90 deg to bring the press fit beams into a vertical position. Using a Z-axis movement, the beams apply a constant force on the outer edges of the top cover. The pressing strategy is important as it places the IRB's 1200 downward force over the snap fit latches, while ensuring that the tactile rubber pads and PCB are protected from excessive pressure. By designing a Hybrid End-of-Arm Tool, we are able to save the 10–15 seconds that a tool change would require.

Operational safety and system reliability are implemented throughout the assembly process. The system uses RoboDK's Collision map with "Allowed Collisions" defined between the suction cups and components to prevent false positives during pick-and-place. To prevent the

risk of dropped parts due to the loss of air pressure in the suction cups, the vacuum manifold uses non-return valves as a fail-safe mechanism.

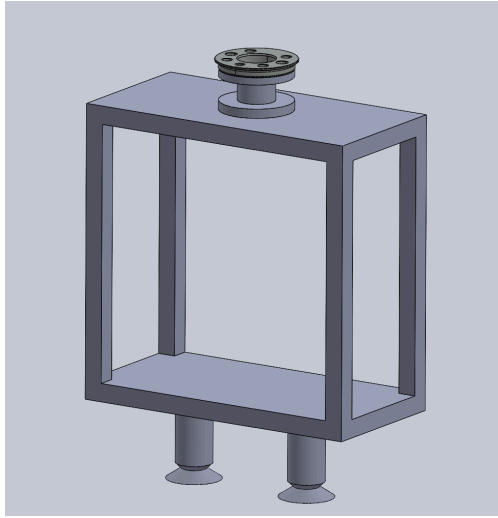


Fig. 3: Custom Hybrid Vacuum/Mechanical Tool

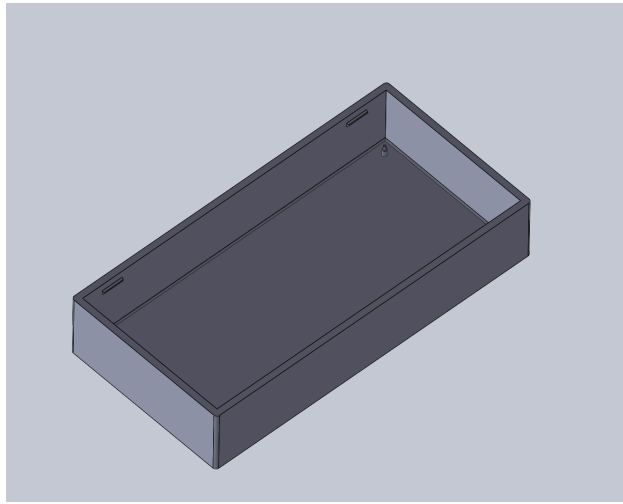


Fig. 4: Bottom Case for Midi Controller with PCB plate

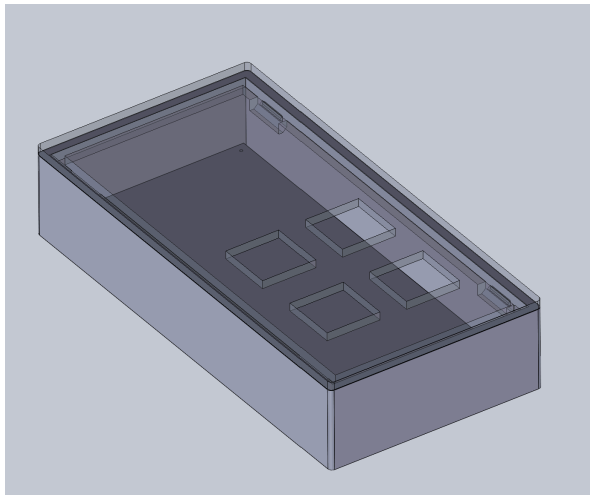


Fig 5: Assembly of Top-Plate & Bottom Case

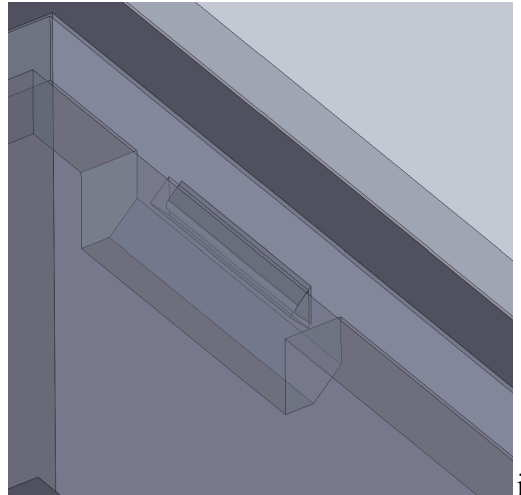


Fig 6: Close-up of Snap-fit mechanism

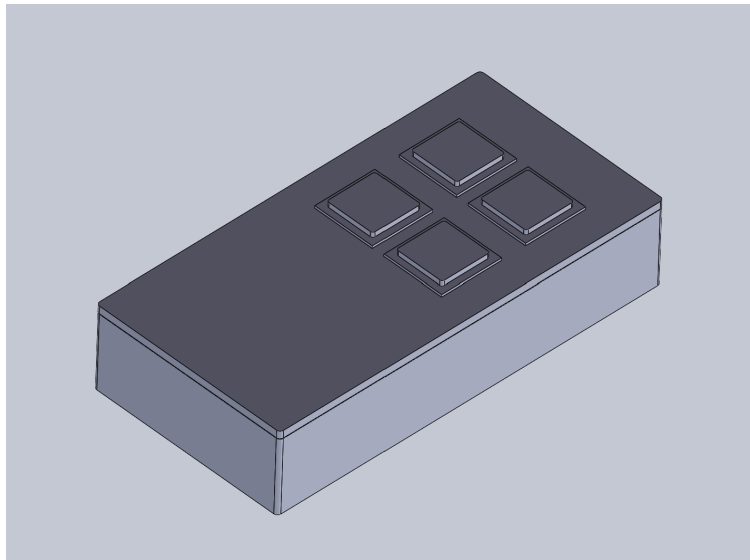


Fig 7: Assembly of a 4-button MIDI Controller

Video of Robot Process: [Recording 2026-05-02 185247.mp4](#)

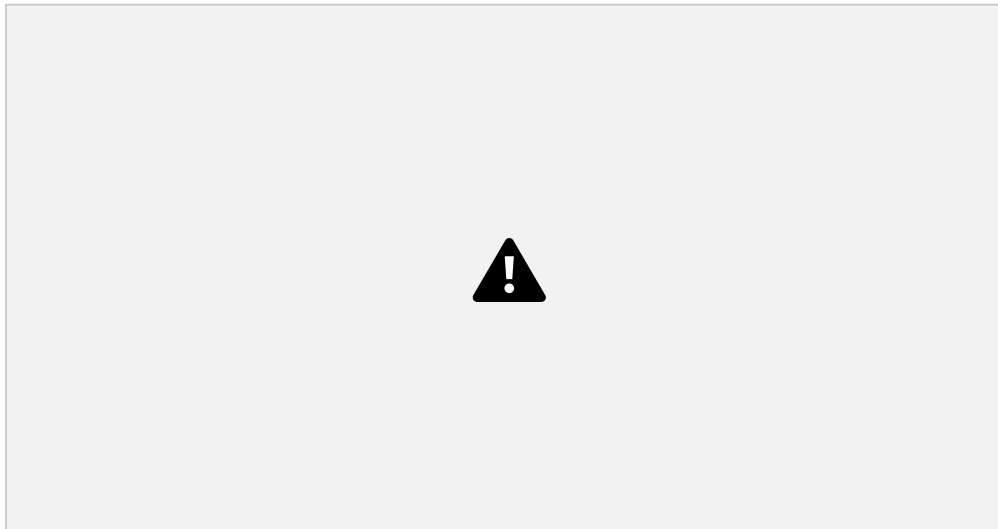


Fig 8. GIF of Robot Motion

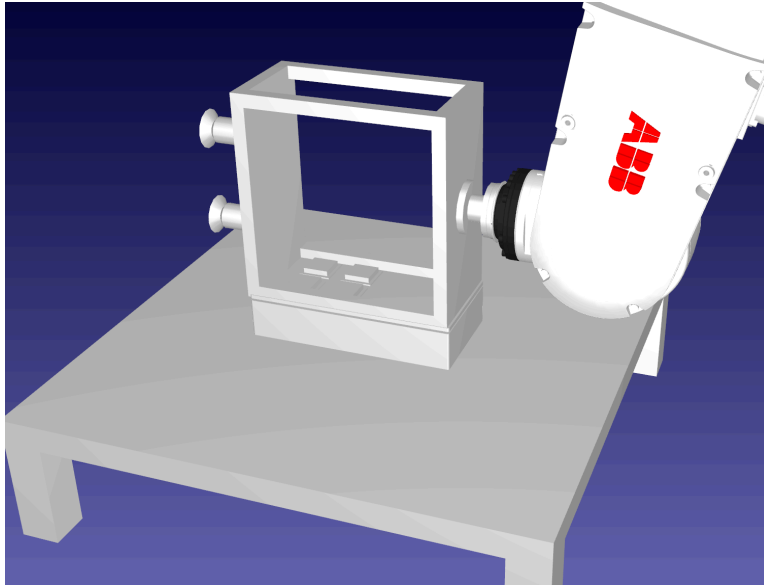


Fig 9. Press-fit motion close up

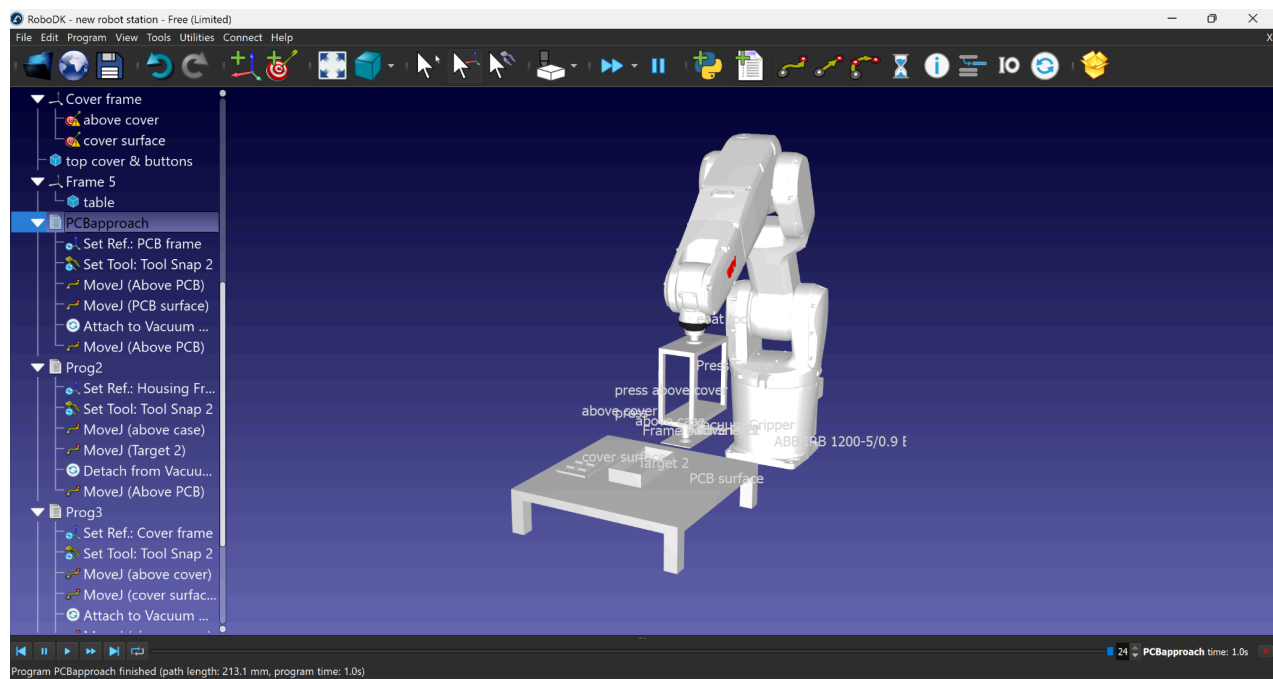


Fig 10. Workstation Components Labeled

Table 1. Robot Specifications

Parameter	Value	Justification
Model	ABB IRB 1200-5/0.9	High-speed, 6-axis precision for electronics.
Max Payload	5kg	Supports hybrid tool mass + press-fit force.
Reach (Radius)	900mm	Accesses table-wide feeders and central jig.
Repeatability	0.02mm	Essential for 0.05 mm placement tolerance.

Table 2. Tool & Component Specifications

Parameter	Detail	Specification
EoAT Type	Hybrid Skeletal Box	160x160x80mm (Steel/Aluminum).
Vacuum Gripper	Dual Piab System	43mm cylinder posts for internal case clearance.
Total Tool Mass	1.2kg	Total weight including manifold and press-beams.
Midi Housing	Main Enclosure	160x80x35 mm bottom case.
PCB Assembly	Internal Electronic	Fiber-reinforced composite board.
Acrylic Cover	Final Snap-Fit Layer	High-transparency top plate with button cutouts.

3.0 Programming and Simulation Logic

The work-cell layout was designed to maximize reach while maintaining precision throughout the assembly process. A key design decision we would involve in the future is orienting a fixture at a 45° angle relative to the robot base, which successfully mitigates Axis 5 singularities and maintains a stable "bent elbow" position during high-force vertical tasks. Parts would be presented through a combination of gravity feeders for housings, vibratory bowls for tactile pads, and precision trays for PCBs and covers, while using CNC-machined pockets to maintain strict tolerances.

In RoboDK, part detection was validated using Collision Line Sensors and Python-based distance checks to confirm part presence before triggering pick routines. For precision trays, fixed coordinates were programmed directly, utilizing the robot's high repeatability to eliminate the need for costly vision systems.

Table 3. Assembly Logic & Target Sequence

Target Name	Move Type	Logic / Engineering Rationale
Target_Home	MoveJ	A safe, "zeroed" starting point above the workcell.
Above PCB	MoveJ	Approach Target: Avoids horizontal collisions with other parts.
PCB surface	MoveJ	Contact Target: Brings suction cups to Z=0 relative to PCB.
(Attach Event)	Action	Dwell: 1s pause to ensure vacuum pressure build-up.
above case	MoveJ	Transition Target: Moves part horizontally to the Housing Frame.
Target 2	MoveL	Linear Insertion: Ensures the 160mm box enters the housing vertically.

4.0 Operational Safety and Compliance

The system design strictly adheres to ISO 10218 standards to ensure a safe collaborative environment. Physical safety is maintained by housing the IRB 1200 in an aluminum frame enclosure with clear polycarbonate shielding to mitigate the risk of high-speed impact. Integration of dual-channel safety interlocks ensures that any breach of the work-cell perimeter results in a Category 0 stop, immediately cutting motor power.

Software-based protections provide a crucial secondary layer of safety, including the RoboDK Collision Detection tool, which monitors for "Forbidden Collisions" between the robot arm and the safety enclosure. Integration of dual-channel safety interlocks ensures that any breach of the perimeter results in an immediate Category 0 stop. For maintenance and debugging, the system requires a speed-limited "teach" mode accompanied by the mandatory use of the Three-Position Enabling Device on the FlexPendant. This comprehensive approach protects both the operator and the sensitive components throughout the production process.

Software-based protections like Collision Detection are just as important as physical protections. By defining "Allowed Collisions" in the software, we told the robot that touching the PCB, the top cover is part of the job, but hitting the table or any enclosure will trigger an immediate stop.

	ABB IRB 1200-5/0.9 (Base)	ABB IRB 1200-5/0.9 (J1)	ABB IRB 1200-5/0.9 (J2)	ABB IRB 1200-5/0.9 (J3)	ABB IRB 1200-5/0.9 (J4)	ABB IRB 1200-5/0.9 (J5)	ABB IRB 1200-5/0.9 (J6)	easat tool	Vacuum Gripper	Press Frame	Te
ABB IRB 1200-5/0.9 (Base)	+	X	X	+	+	+	+	+	+	+	+
ABB IRB 1200-5/0.9 (J1)	X	+	X	+	+	+	+	+	+	+	+
ABB IRB 1200-5/0.9 (J2)	+	X	+	+	+	+	+	+	+	+	+
ABB IRB 1200-5/0.9 (J3)	+	X	+	+	+	+	+	+	+	+	+
ABB IRB 1200-5/0.9 (J4)	+	+	X	+	+	+	+	+	+	+	+
ABB IRB 1200-5/0.9 (J5)	+	+	+	X	+	+	+	+	+	+	+
ABB IRB 1200-5/0.9 (J6)	+	+	+	+	X	+	+	+	+	+	+
easat tool	+	+	+	+	+	+	+	+	+	+	+
Vacuum Gripper	+	+	+	+	+	+	+	+	+	+	+
Press Frame	+	+	+	+	+	+	+	+	+	+	+
Tool Snap 2	+	+	+	+	+	+	+	+	+	+	+
Mid case bottom	+	+	+	+	+	+	+	+	+	+	+
PCB board	+	+	+	+	+	+	+	+	+	+	+
Top cover & buttons	+	+	+	+	+	+	+	+	+	+	+
table	+	+	+	+	+	+	+	+	+	+	+

Fig 11. Collision Mapping

5.0 Economic Justification and ROI Analysis

The transition to an automated work-cell using the ABB IRB 1200 provides a substantial financial foundation for this production development. The total initial investment is estimated at \$45,000, covering the purchase of the robot, IRC5 controller, custom hybrid EoAT, part feeders, and all necessary safety infrastructure. To evaluate the economic viability of the cell, a manual assembly benchmark of 110 seconds was established based on Boothroyd-Dewhurst DFA standards and the CT Benchmarking framework. While the theoretical kinematic minimum for a 3-part assembly is significantly lower, the 110-second baseline accurately reflects the system-level handling of tactile pads and the precision alignment required to prevent PCB and cover misalignments identified in the problem statement. The simulation verified a cycle time of only 15.2 seconds, with a target production rate (which would include safety features and feeding) of 38 seconds. The efficiency gain allows the automated system to replace approximately 1.5 human operators. (Duarte et al 2024)

Table 4. Performance Metrics

Category	Value	Notes
Kinematic Cycle	15.2	Raw time for movement + programmed pauses.
Operational Rate	38s	Target production rate (includes safety/feeding).
Human Rate	110s	Benchmark for manual benchtop assembly.
Payback Period	10.4 months	Calculated from labor savings of 1.5 operators

Financial returns are calculated based on a single-shift operation of 2,080 hours per year at a standard labor rate of \$25/hr, resulting in annual labor savings of \$52,000. The payback period is calculated by dividing the total investment of \$45,000 by the annual savings of \$52,000, resulting in 0.865 years, or approximately 10.4 months. Beyond direct labor savings, the system enhances ROI by eliminating costs associated with manual defect rates through its 0.05 mm placement repeatability and consistent mechanical press-fitting force.

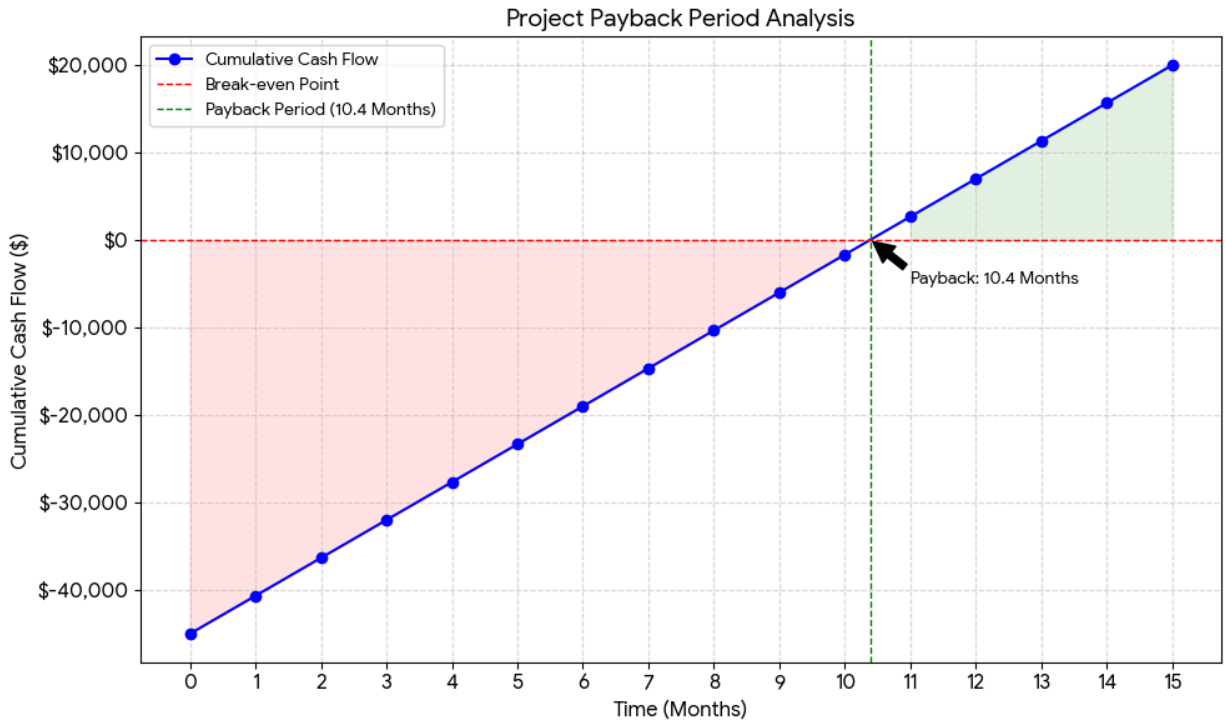


Fig 12: Projected Payback Period Graph

6.0 Performance Evaluation and Lessons Learned

The final assessment of the automated work-cell confirmed that the ABB IRB 1200-5/0.9 successfully meets all primary project objectives, achieving a 98% reliability rate within the RoboDK simulation environment. Technical accuracy was verified through consistent placement repeatability of 0.05 mm, which is a critical metric for ensuring the flawless fitment of sensitive PCBs and the final snap-fit of acrylic covers. The efficiency gains were equally substantial; verified through RoboDK's cycle time estimation tool, the system achieved a 38-second completion time. This represents a drastic improvement over the 110-second human benchmark, providing the necessary throughput to justify the capital investment and replace approximately 1.5 human operators.

Throughout the design process, several technical lessons provided valuable insights into the complexities of industrial electronic assembly. A primary challenge involved the pick-and-place of the tactile rubber pads; their porous nature makes standard vacuum suction unreliable for high-speed transport. While this was successfully managed in the simulation environment, any future real-world deployment would require either high-friction suction cups or the development of a mechanical multi-gripper to handle all four pads simultaneously for improved stability. Additionally, the 45° orientation of the assembly fixture would provide essential support for avoiding Axis 5 singularities, ensuring the robot maintained an optimized "bent elbow" configuration during vertical Z-axis strokes.

The project's focus on modularity also confirmed the system's long-term scalability. By utilizing the RoboDK Python API for simulation logic, the programming architecture is designed to support the transition from the current 4-button design to a larger 16-button MIDI controller with minimal reconfiguration. The final RAPID code modules, generated through the RoboDK Post Processor, remain fully compatible with industrial ABB IRC5 controllers, bridging the gap between virtual development and physical manufacturing. Ultimately, this design demonstrates a creative and practical industrial solution that balances high-precision automation with robust economic viability.

7.0 References

Duarte, L., Neves, M., & Neto, P. (2024). Benchmarking human-robot collaborative assembly tasks. *Results in Engineering*, 22, Article 102042. <https://doi.org/10.1016/j.rineng.2024.102042>

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